Notes and Research in Anomaly Mediated Supersymmetry Breaking (AMSB):

Structural Aspects and Vacuum Stabilization of ASQCD

1. Anomaly mediated SUSY breaking (AMSB) from superconformal (SCF) anomaly

The crucial feature of the scenario we consider is that SUSY breaking effects in the observable sector have a pure gravitational origin. This is equivalent to saying that the only relevant source of soft terms is the auxiliary field u of the off-shell gravitational supermultiplet for Poincare supergravity (SUGRA). The natural formalism to describe these effects is the SCF calculus formulation of SUGRA.

One start from SCF SUGRA. A 4D N=1 SUGRA can be seen as: SCF + gauge fixing, whose point is to introduce a compensator superfield $\Phi = 1 + \theta^2 F_\Phi$ which is a chiral superfield. Under super-Weyl transformation $\Phi \to e^{-2\sigma}\Phi$, it counts Weyl weight and make action manifestly superconformal.

In classical theory, this compensator can be cancelled, with full SUSY and SCF (and QCD symmetries). While once quantized, the theory can no longer contain such much symmetries. If one keeps SCF, it will cause anomaly in SUSY and gives a SUSY breaking. In classical theory, consider a kinetic term $\int d^4\theta \Phi^\dagger \Phi Q^\dagger Q$ and superpotential $\int d^2\theta \Phi^3 W(Q)$. Under rescaling $Q \to \Phi^{-1}Q$, $\Phi^\dagger \Phi Q^\dagger Q \to Q^\dagger Q$, in which the compensator is cancelled. While in quantum theory, the measure of path integral $[DQ]$ is not invariant under the rescaling and Jacobian gives $\ln\ \Phi W^\alpha W_\alpha$, which is super-Weyl anomaly.

Consider a classical gauge kinetic term $\int d^2\theta \frac{1}{4g^2} W^\alpha W_\alpha$ that do not include Φ. Quantize it gives $g = g(\mu)$ that has RG running. In SCF, the scale μ has to be Weyl invariant $\mu \to \frac{\mu}{\Phi}$, thus, $g(\mu) \to g\left(\frac{\mu}{\Phi}\right)$. Note that $\Phi = 1 + \theta^2 F_\Phi$,

$$ln\frac{\mu}{\Phi} = ln\,\mu - \theta^2 F_\Phi,$$

$$\frac{1}{g^2(\mu/\Phi)} = \frac{1}{g^2(\mu)} - \frac{\partial}{\partial \ln\ \mu}\frac{1}{g^2}\theta^2 F_\Phi.$$

Using $\beta(g) = \frac{dg}{d\ln\ \mu}$, one gives

$$\frac{1}{g^2(\mu/\Phi)} = \frac{1}{g^2} + \frac{2\beta(g)}{g^3}\theta^2 F_\Phi,$$

Which corresponds to gaugino mass

$$\boldsymbol{M_\lambda = \frac{\beta(g)}{g} F_\Phi}.$$

Consider matter sector. Wilson effective action: $\int d^4\theta Z_i(\mu) Q_i^\dagger Q_i$, in super-Weyl invariant

form, $Z_i\left(\frac{\mu}{\sqrt{\Phi^\dagger\Phi}}\right)$.

Define anomalous dimension: $\gamma_i = \frac{d\ln\ Z_i}{d\ln\ \mu}$, using $ln\left(\Phi^\dagger\Phi\right) = \theta^2 F_\Phi + \bar{\theta}^2 F_\Phi^\dagger + \cdots$, thus $Z_i = Z_i(\mu) - \gamma_i Z_i(\theta^2 F_\Phi + \bar{\theta}^2 F_\Phi^\dagger) + \cdots$.

Superpotential $W = y_{ijk} Q_i Q_j Q_k$, after canonical normalization, Yukawa coupling

$$y_{ijk}^{\text{phys}} \propto \frac{y_{ijk}}{\sqrt{Z_i Z_j Z_k}},$$

thus

$$\boldsymbol{A_{ijk} = -(\gamma_i + \gamma_j + \gamma_k) F_\Phi}.$$

Look at the next order of the expansion of $Z_i\left(\frac{\mu}{\sqrt{\Phi^\dagger\Phi}}\right)$ gives $\theta^2\bar{\theta}^2|F_\Phi|^2$.

$$\boldsymbol{m_i^2 = -\frac{1}{4}|F_\Phi|^2 \frac{d\gamma_i}{d\ln\ \mu}},$$

In which $\frac{d\gamma_i}{d\ln\ \mu} = \beta_g \frac{\partial\gamma_i}{\partial g} + \beta_y \frac{\partial\gamma_i}{\partial y}$.

2. Renormalization group (RG) flow

Renormalization Group (RG) flow describes how renormalized couplings depend on the renormalization scale μ. In quantum field theory:

$$\mu\frac{d}{d\mu}(\text{renormalized parameters}) \neq 0.$$

Define beta function

$$\beta_g \equiv \mu\frac{dg}{d\mu}$$

and anomalous dimension

$$\gamma_i \equiv \mu\frac{d\ln\ Z_i}{d\mu},$$

where Z_i is the wavefunction renormalization of field Q_i.

It is often convenient to define $t = \ln\ \mu$, so RG equations become:

$$\frac{dg}{dt} = \beta_g, \frac{d\ln\ Z_i}{dt} = \gamma_i.$$

Start from bare vs renormalized quantities. Fields $Q_{0i} = Z_i^{1/2}(\mu)\ Q_i(\mu)$. Since the bare field is independent of μ, $0 = \mu\frac{d}{d\mu}Q_{0i}$. This implies

$$\mu\frac{dQ_i}{d\mu} = -\frac{1}{2}\gamma_i Q_i.$$

So RG flow of fields is governed entirely by γ_i.

Similarly for couplings $g_0 = Z_g(g,\mu)\ g(\mu)$. Differentiating at fixed bare coupling gives

$$\beta_g = \mu\frac{dg}{d\mu}.$$

3. Algebraic and geometric structure of AMSB

Rigid SUSY on curved space means there exists spinor parameter $\epsilon(x)$ s.t. background gravitino variation is 0. In SUGRA, $\delta\psi_\mu = \nabla_\mu\epsilon + \cdots$. For AdS background, auxiliary field/background flux gives

$$\delta\psi_\mu = \nabla_\mu\epsilon - \frac{i}{2L}\gamma_\mu\epsilon.$$

Note that preserved SUSY requires

$$\delta\psi_\mu = 0,$$

thus

$$\boxed{\nabla_\mu\epsilon = \frac{i}{2L}\gamma_\mu\epsilon}$$

Majorana conjugate gives

$$\nabla_\mu\bar{\epsilon} = -\kappa\,\bar{\epsilon}\,\gamma_\mu,$$

In which $\kappa = \frac{i}{2L}$.

(Covariant derivative on Killing spinor

$$[\nabla_\mu, \nabla_\nu]\epsilon = \nabla_\mu(\kappa\gamma_\nu\epsilon) - \nabla_\nu(\kappa\gamma_\mu\epsilon) = \kappa^2(\gamma_\nu\gamma_\mu - \gamma_\mu\gamma_\nu)\epsilon = 2\kappa^2\gamma_{\mu\nu}\epsilon.$$

Note that

$$[\nabla_\mu, \nabla_\nu]\epsilon = \frac{1}{4}R_{\mu\nu\rho\sigma}\gamma^{\rho\sigma}\epsilon,$$

and for AdS_4,

$$R_{\mu\nu\rho\sigma} = -\frac{1}{L^2}(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho}),$$

thus

$$[\nabla_\mu, \nabla_\nu]\epsilon = -\frac{1}{2L^2}\gamma_{\mu\nu}\epsilon.$$

Compare with $\kappa^2 = -1/(4L^2)$, so

$$2\kappa^2 = -\frac{1}{2L^2}.)$$

Define a bilinear

$$\xi^\mu \equiv \bar{\epsilon}_2\gamma^\mu\epsilon_1.$$

Its covariant derivative

$$\nabla_\mu\xi_\nu = (\nabla_\mu\bar{\epsilon}_2)\gamma_\nu\epsilon_1 + \bar{\epsilon}_2\gamma_\nu\nabla_\mu\epsilon_1.$$

Plug in Killing spinor equation

$$\nabla_\mu\xi_\nu = -\kappa\bar{\epsilon}_2\gamma_\mu\gamma_\nu\epsilon_1 + \kappa\bar{\epsilon}_2\gamma_\nu\gamma_\mu\epsilon_1 = \kappa\bar{\epsilon}_2[\gamma_\nu, \gamma_\mu]\epsilon_1 = 2\kappa\,\bar{\epsilon}_2\gamma_{\nu\mu}\epsilon_1.$$

Thus

$$\nabla_{(\mu}\xi_{\nu)} = 0.$$

So ξ^μ is a Killing vector.

Choose a standard curved background rigid SUSY transform on field (ϕ, ψ, F) (that close under $\nabla_\mu\epsilon = \kappa\gamma_\mu\epsilon$)

$$\delta_\epsilon\phi = \bar{\epsilon}\psi,$$

$$\delta_\epsilon \psi = \gamma^\mu \epsilon \nabla_\mu \phi + F\epsilon,$$
$$\delta_\epsilon F = \bar{\epsilon}\gamma^\mu \nabla_\mu \psi - 2\kappa\, \bar{\epsilon}\psi.$$

On scalar field,

$$[\delta_1, \delta_2]\phi = \bar{\epsilon}_2\, \delta_1\psi - \bar{\epsilon}_1\, \delta_2\psi,$$
$$[\delta_1, \delta_2]\phi = (\bar{\epsilon}_2\gamma^\mu\epsilon_1 - \bar{\epsilon}_1\gamma^\mu\epsilon_2)\nabla_\mu\phi.$$

Define

$$\xi^\mu \equiv \bar{\epsilon}_2\gamma^\mu\epsilon_1 - \bar{\epsilon}_1\gamma^\mu\epsilon_2,$$

thus

$$[\delta_1, \delta_2]\phi = \xi^\mu\nabla_\mu\phi.$$

On spinor field,

$$[\delta_1, \delta_2]\psi = \delta_1(\gamma^\mu\epsilon_2\nabla_\mu\phi + F\epsilon_2) - (1 \leftrightarrow 2).$$

One gives

$$[\delta_1, \delta_2]\psi = \xi^\mu\nabla_\mu\psi + \frac{1}{4}\,\Omega^{ab}\gamma_{ab}\psi,$$

(Kosmann Lie derivative) in which

$$\Omega^{ab} = \nabla^{[a}\xi^{b]} = 2\kappa(\bar{\epsilon}_2\gamma^{ab}\epsilon_1 - \bar{\epsilon}_1\gamma^{ab}\epsilon_2).$$

Here $\frac{1}{4}\,\Omega^{ab}\gamma_{ab}$ term is from accompanying Lorentz rotation.

Define:

- Q_α: supercharge； ;
- P_a: translation-like generator； ;
- M_{ab}: Lorentz generator;

SUSY transformation:

$$\delta_\epsilon = \bar{\epsilon}Q = \bar{\epsilon}^\alpha Q_\alpha,$$

$$[\delta_{\epsilon_1}, \delta_{\epsilon_2}] = \bar{\epsilon}_2^\alpha\bar{\epsilon}_1^\beta\{Q_\alpha, Q_\beta\}.$$

On the other hand,

$$[\delta_{\epsilon_1}, \delta_{\epsilon_2}] = \xi^a P_a + \frac{1}{2}\Omega^{ab}M_{ab},$$

where

$$P_a = e_a{}^\mu\nabla_\mu,$$

thus

$$\bar{\epsilon}_2^\alpha\bar{\epsilon}_1^\beta\{Q_\alpha, Q_\beta\} = \xi^a P_a + \frac{1}{2}\Omega^{ab}M_{ab}.$$

$$\xi^a = \bar{\epsilon}_2\gamma^a\epsilon_1 - \bar{\epsilon}_1\gamma^a\epsilon_2,$$

using

$$\bar{\epsilon}_1\gamma^a\epsilon_2 = -\bar{\epsilon}_2\gamma^a\epsilon_1,$$

thus

$$\xi^a = 2\bar{\epsilon}_2\gamma^a\epsilon_1,$$

gives the translation term

$$\xi^a P_a = 2(\bar{\epsilon}_2\gamma^a\epsilon_1)P_a.$$

So supercharge anticommutator includes

$$\{\boldsymbol{Q_\alpha}, \boldsymbol{\bar{Q}_{\dot\beta}}\} = \boldsymbol{2(\gamma^a)_{\alpha\dot\beta} P_a}.$$

$$\Omega^{ab} = \frac{i}{L}(\bar\epsilon_2\gamma^{ab}\epsilon_1 - \bar\epsilon_1\gamma^{ab}\epsilon_2),$$

using

$$\bar\epsilon_1\gamma^{ab}\epsilon_2 = -\bar\epsilon_2\gamma^{ab}\epsilon_1,$$

thus

$$\Omega^{ab} = \frac{2i}{L}\bar\epsilon_2\gamma^{ab}\epsilon_1,$$

$$\frac{1}{2}\Omega^{ab}M_{ab} = \frac{i}{L}(\bar\epsilon_2\gamma^{ab}\epsilon_1)M_{ab}.$$

Note that

$$\gamma^{ab} = \frac{1}{2}[\gamma^a, \gamma^b]$$

is symmetric on spinor indices, so it corresponds to

$$\{Q_\alpha, Q_\beta\}.$$

Thus

$$\{\boldsymbol{Q_\alpha}, \boldsymbol{Q_\beta}\} = \boldsymbol{\frac{i}{L}(\gamma^{ab}C^{-1})_{\alpha\beta} M_{ab}},$$

and its Majorana conjugate

$$\{\boldsymbol{\bar{Q}_{\dot\alpha}}, \boldsymbol{\bar{Q}_{\dot\beta}}\} = -\boldsymbol{\frac{i}{L}(C^{-1}\gamma^{ab})_{\dot\alpha\dot\beta} M_{ab}}.$$

(AdS deformation)

(Charge conjugation matrix

Supercharge Q_α is spinor, so $\{Q_\alpha, Q_\beta\}$ has two spinor indices, so as right side. While M_{ab} only has Lorentz indices, thus needs an intertwiner $(\gamma^{ab}C^{-1})_{\alpha\beta}$.

Define charge conjugation matrix

$$C\gamma^a C^{-1} = -(\gamma^a)^T$$

or

$$(\gamma^a)^T = -C\gamma^a C^{-1}.$$

Not that $(\gamma^{ab})_\alpha{}^\beta$ has an upper index, so use C^{-1} to lower it

$$(\gamma^{ab}C^{-1})_{\alpha\beta}.$$

In Majorana basis $C = \gamma^0$, in Dirac basis $C = i\gamma^2\gamma^0$ which satisfies $C^T = -C$.)

Jacobi identity

$$[P_a, \{Q_\alpha, \bar{Q}_{\dot\beta}\}] = \{[P_a, Q_\alpha], \bar{Q}_{\dot\beta}\} + \{Q_\alpha, [P_a, \bar{Q}_{\dot\beta}]\}.$$

Since SUSY generator is characterized by Killing spinor, translation acting on SUSY parameter $[P_a, Q]$ corresponds to $\nabla_a\epsilon$. Thus

$$\boldsymbol{[P_a, Q_\alpha] = \frac{i}{2L}(\gamma_a)_\alpha{}^\beta Q_\beta},$$

$$[\boldsymbol{P_a}, \bar{\boldsymbol{Q}}_{\dot{\alpha}}] = -\frac{\boldsymbol{i}}{\boldsymbol{2L}}(\boldsymbol{\gamma_a})_{\dot{\alpha}}{}^{\dot{\beta}}\bar{\boldsymbol{Q}}_{\dot{\beta}}.$$

Plug in $\{Q_\alpha, \bar{Q}_{\dot{\beta}}\} = 2(\gamma^c)_{\alpha\dot{\beta}}P_c$, LHS gives

$$\left[P_a, 2(\gamma^c)_{\alpha\dot{\beta}}P_c\right] = 2(\gamma^c)_{\alpha\dot{\beta}}[P_a, P_c].$$

RHS:

$$\{[P_a, Q_\alpha], \bar{Q}_{\dot{\beta}}\} = \frac{i}{2L}(\gamma_a)_\alpha{}^\gamma\{Q_\gamma, \bar{Q}_{\dot{\beta}}\} = \frac{i}{L}(\gamma_a\gamma^c)_{\alpha\dot{\beta}}P_c,$$

$$\{Q_\alpha, [P_a, \bar{Q}_{\dot{\beta}}]\} = -\frac{i}{L}(\gamma^c\gamma_a)_{\alpha\dot{\beta}}P_c,$$

gives $\mathrm{RHS} = \frac{i}{L}[\gamma_a, \gamma^c]_{\alpha\dot{\beta}}P_c = \frac{2i}{L}(\gamma_{ac})_{\alpha\dot{\beta}}P^c$.

Using Lorentz algebra $[M_{ab}, P_c] = \eta_{bc}P_a - \eta_{ac}P_b$, one gives

$$[\boldsymbol{P_a}, \boldsymbol{P_b}] = -\frac{\boldsymbol{1}}{\boldsymbol{L^2}}\boldsymbol{M_{ab}}.$$

Taken together:

$$\boxed{\{Q_\alpha, \bar{Q}_{\dot{\beta}}\} = 2(\gamma^a)_{\alpha\dot{\beta}}P_a}$$

$$\boxed{\{Q_\alpha, Q_\beta\} = \frac{i}{L}(\gamma^{ab}C^{-1})_{\alpha\beta}M_{ab}}$$

$$\boxed{[P_a, Q_\alpha] = \frac{i}{2L}(\gamma_a)_\alpha{}^\beta Q_\beta}$$

$$\boxed{[P_a, P_b] = -\frac{1}{L^2}M_{ab}}$$

with ordinary Lorentz algebra:

$$[M, M] \sim M, [M, P] \sim P, [M, Q] \sim Q,$$

gives Lie superalgebra of $OSp(1 \mid 4)$

(Flat limit:

Let

$$L \to \infty,$$

gives

$$\{Q, Q\} \to 0,$$
$$[P_a, Q] \to 0,$$
$$[P_a, P_b] \to 0.$$

with

$$\{Q_\alpha, \bar{Q}_{\dot{\beta}}\} = 2(\gamma^a)_{\alpha\dot{\beta}}P_a,$$

ordinary super-Poincaré algebra.)

Now promote rigid SUSY to local SUSY. We introduce the chiral conformal compensator superfield: $\Phi = 1 + \theta^2 F_\Phi$. In SUGRA gauge fixing:

$$F_\Phi = m_{3/2} + \frac{1}{3}K_i F^i,$$

in the sequestered limit (AMSB case)

$$F_\Phi = m_{3/2}.$$

Now we identify the AdS scale

$$\frac{1}{L} \equiv m_{3/2}.$$

So the Killing spinor equation becomes

$$\nabla_\mu \epsilon = \frac{i}{2} m_{3/2} \gamma_\mu \epsilon.$$

In SUGRA, consistency with local SUSY forces $\mu \to \frac{\mu}{\Phi}$ since Weyl invariance: μ has Weyl weight 1, Φ has Weyl weight 1, so only μ/Φ is invariant. Thus every coupling becomes a superfield:

$$g \to g\left(\frac{\mu}{\Phi}\right), Z_i \to Z_i\left(\frac{\mu}{\Phi}, \frac{\mu}{\Phi^\dagger}\right)$$

Rigid SUSY on curved space is more general than AdS. Festuccia and Seiberg show that one can construct supersymmetric theories on S^4, $S^3 \times \mathbb{R}$, and other backgrounds by turning on appropriate auxiliary or background gauge fields. But these cases are not the same as the AMSB/AdS setup: S^4 is Euclidean and, unless the theory is conformal, the resulting Lagrangian is not reflection positive; $S^3 \times \mathbb{R}$ requires a continuous R-symmetry and additional background gauge structure. By contrast, the AdS branch comes directly from the old-minimal supergravity auxiliary fields with a single intrinsic scale and no extra background gauge fields in the maximally symmetric branch. Only AdS_4 gives the maximally symmetric Lorentzian SUSY vacuum whose unique scale matches $m_{3/2}$ and the compensator in AMSB. [1105.0689]

4. Run-away behavior of supersymmetric QCD (SQCD) and vacuum stabilization by anomaly mediation (ASQCD)

It has been a long-standing goal in theoretical physics to understand strongly correlated systems. Well-known examples are high-temperature superconductors in condensed matter physics and strong interaction in particle physics. The latter is described by Quantum Chromo Dynamics (QCD) based on SU(3) gauge theory with quarks and gluons as fundamental degrees of freedom. At high energies, its dynamics can be studied with perturbation theory thanks to asymptotic freedom. On the other hand, at low energies, theory becomes strongly coupled at a scale Λ that makes fundamental degrees of freedom trapped inside bound states, and typical coupling con stants among bound states are ~ O(4π) beyond the reach of perturbation theory. In particular, QCD-like theories with massless quarks are believed to dynamically break the chiral symmetry which has been difficult to demonstrated analytically.

Certain strongly coupled theories have exact solutions in the infrared (IR). The important example for the discussion here is supersymmetric QCD (SQCD) whose dynamics was worked out by Seiberg. Yet SQCD without the quark mass terms has extra massless degrees freedom, namely squarks and gauginos, which are not present in QCD. No limit

of SQCD seems to mimic the dynamics of QCD. One can introduce supersymmetry breaking to SQCD to mimic non-supersymmetric QCD. In many cases, there is no full theoretical control, or a sign that there is a phase transition when supersymmetry breaking becomes as large as the dynamical scale. AMSB is an exception. It has a property called ultraviolet (UV) insensitivity, and the supersymmetry breaking effects are completely controlled at all energy scales with any description. Therefore, the known exact results in SQCD can be generalized to include the effects of AMSB. In addition, ASQCD makes both squarks and gauginos massive, and hence its massless particle content is identical to that of QCD. Therefore, ASQCD is a QCD-like theory.

In the supersymmetric SU(Nc) QCD with Nf flavors in the fundamental representation (SQCD), one can give the leading order of ASQCD

$$m_Q^2 = m_{\tilde{Q}}^2 = \frac{g^4}{(8\pi^2)^2} 2C_i(3N_c - N_f)m^2,$$
$$m_\lambda = \frac{g^2}{16\pi^2}(3N_c - N_f)m.$$

Here, $C_i = \frac{N_c^2-1}{2N_c}$, and $m_Q^2 > 0$ in the range Nf < 3Nc where the theory is asymptotically free. Therefore, the squarks and gauginos acquire mass and the massless par ticle content is identical to that of non-supersymmetric QCD. As m is increased beyond the dynamical scale of the gauge theory Λ, gluinos and squarks can be integrated out, and the theory does become QCD.

It is an open problem whether the change in m is continuous. [I should look at it.] Nonetheless, SQCD with AMSB is continuously connected to QCD.

The most remarkable property of the anomaly mediated supersymmetry breaking is its **ultraviolet-insensitivity**. The expressions for the supersymmetry breaking parameters above depend on wave function renormalization and running coupling constants, which jump when heavy fields are integrated out from the theory. It turns out that the threshold corrections from the loops of heavy fields precisely give the necessary jump. Therefore the above expressions remain true at all energy scales and depend only on the particle content and interactions present at that energy scale.

Assume $N_c \geq 3$.

$$N_f < N_c$$

Non-perturbative Affleck–Dine–Seiberg (ADS) superpotential

$$W = (N_c - N_f)\left(\frac{\Lambda^{3N_c-N_f}}{\det M}\right)^{1/(N_c-N_f)},$$

where meson fields M^{ij}.

The SQCD has a run-away potential and hence no ground states. When $M \gg \Lambda^2, M_{ij} = M\delta_{ij}$ describes the D-flat direction

$$Q = \tilde{Q} = \begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \\ 0 & \cdots & 0 \\ \vdots & \vdots & \vdots \\ 0 & \cdots & 0 \end{pmatrix} \phi, M = \phi^2.$$

The upper part is $N_f \times N_f$ and lover part $(N_c - N_f) \times N_f$. One gives the Lagrangian along this direction in ASQCD

$$\mathcal{L} = \int d^4\theta \Phi^* \Phi 2N_f \phi^* \phi + \int d^2\theta \Phi^3 (N_c - N_f) \left(\frac{\Lambda^{3N_c - N_f}}{\phi^{2N_f}} \right)^{1/(N_c - N_f)}$$

And the corresponding potential

$$\mathrm{V} = \left| 2N_f \frac{1}{\phi} \left(\frac{\Lambda^{3N_c - N_f}}{\phi^{2N_f}} \right)^{1/(N_c - N_f)} \right|^2 - (3N_c - N_f) m \left(\frac{\Lambda^{3N_c - N_f}}{\phi^{2N_f}} \right)^{1/(N_c - N_f)} + c.c..$$

Now there is a well-defined minimum $M_{ij} = \Lambda^2 \left(\frac{4N_f(N_c + N_f)}{3N_c - N_f} \frac{\Lambda}{m} \right)^{(N_c - N_f)/N_c} \delta_{ij} \gg \Lambda^2$.

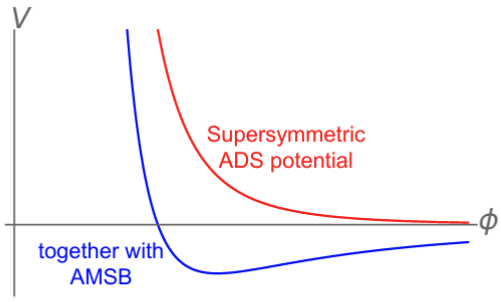

Figure 1: Schematics of the potential for $N_f < N_c$. The red curve is for SQCD with run-away behavior, while the blue curve for ASQCD has a well-defined minimum.

The mass for mesons from AMSB is loop suppressed and hence can be ignored.
The $SU(N_f)_Q \times SU(N_f)_{\tilde{Q}}$ flavor symmetry is dynamically broken to $SU(N_f)_V$.
The massless particle spectrum is the corresponding Nambu–Goldstone bosons (pions).
The scalar and fermion partners of the Nambu–Goldstone bosons (NGBs) have mass that grows with m. Naively increasing m beyond Λ, the only remaining degrees of freedom are massless NGBs.
There is no sign of a phase transition and the two limits are likely continuously connected.

$$N_f = N_c$$

Superpotential

$$W = X\left(\det M - \tilde{B}B - \Lambda^{2N_c}\right).$$

By going to canonical normalization of the fields, one gives the superpotential

$$W = X\left(\lambda \frac{\det M}{\Lambda^{N_c - 2}} - \kappa \tilde{B}B - \Lambda^2\right).$$

λ, κ are dimensionless coupling constants. One can find two candidate ground states.

a. $M^{ij} \& = \lambda^{-1/N_c} \Lambda \delta^{ij},\ B = \tilde{B} = 0,\ X = \lambda^{-2/N_c} m,\ V = -N_c \lambda^{-2/N_c} m^2 \Lambda^2$

The massless spectrum is the NGBs of $SU(N_f)_Q \times SU(N_f)_{\tilde{Q}}/SU(N_f)_V$.

b. $M^{ij} = 0,\ B = \tilde{B} = \kappa^{-1/2}\Lambda,\ X = \kappa^{-1}m,\ V = -(2\kappa)^{-1}m^2\Lambda^2$

The massless spectrum is the NGB of spontaneously broken $U(1)_B$ andmesinos that match the anomalies of $SU(N_f)_Q \times SU(N_f)_{\tilde{Q}}$.

Rely on the naive dimensional analysis which suggests $\lambda \approx (4\pi)^{N_c/2}$ and $\kappa \approx 4\pi$, one obtains the minimum of state a: $V \approx -N_c \frac{1}{4\pi} m^2\Lambda^2$, and b: $V \approx -\frac{1}{8\pi} m^2\Lambda^2$. Therefore a is the global minimum, where the massless NGBs of $SU(N_f)_Q \times SU(N_f)_{\tilde{Q}}/SU(N_f)_V$ w/ $f_\pi \approx \Lambda/(4\pi)^2$, as well as baryons that acquire mass $m_B \approx m$.

It shows that QCD with 3 colors and 3 flavors breaks chiral symmetry with massless pions and massive baryons.

$$N_f = N_c + 1$$

It is s-confining w/ superpotential

$$W = \frac{1}{\Lambda^{2N_f-3}}\left(\det M - \tilde{B}_i M^{ij} B_j\right).$$

By going to canonical normalization of the fields, one gives the superpotential

$$W = \lambda \frac{\det M}{\Lambda^{N_f-3}} - \kappa \tilde{B} M B.$$

The corresponding potential for ASQCD for the direction $M^{ij} = \phi\delta^{ij}$ is

$$V = N_f \lambda^2 \frac{|\phi|^{2N_f-2}}{\Lambda^{2N_f-6}} - \lambda(N_f - 3)m\phi^{N_f} + c.c.$$

w/ a minimum at

$$\phi = \Lambda\left(\frac{(N_f-3)m}{(N_f-1)\lambda\Lambda}\right)^{1/(N_f-2)} \ll \Lambda,$$

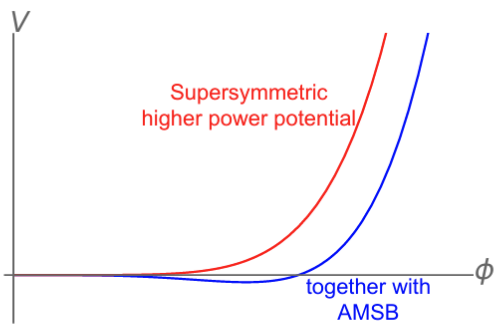

Figure 2: Schematics of the potential for s-confining and free magnetic phases. The red curve is for SQCD with power-law behavior, while the blue curve for ASQCD has a well-defined minimum.

The mass for mesons from AMSB is loop suppressed and hence can be ignored.

The vacuum energy is $V \sim -O(m^{(2N_f-2)}\Lambda^{(2N_f-6)})^{1/(N_f-2)}$.

The massless spectrum is once again the NGBs of $SU(N_f)_Q \times SU(N_f)_{\tilde{Q}}/SU(N_f)_V$.

There is no sign of phase transition as m is increased beyond Λ.

$$N_c + 2 \leq N_f \leq \frac{3}{2} N_c$$

This range is in the **free magnetic** phase described by an $SU(N_f - N_c)$ SQCD as the low-energy theory below the dynamical scale Λ with the superpotential

$$W = \kappa \tilde{q}_i M^{ij} q_j,$$

Where $q(\tilde{q})$ are dual (anti-) quarks. The mesons M are canonically normalized. Similarly to the s-confining case, there is one-loop tri-linear coupling that allows for a minimum with

$$q = \tilde{q} = \begin{pmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \\ 0 & \cdots & 0 \\ \vdots & \vdots & \vdots \\ 0 & \cdots & 0 \end{pmatrix} \phi, M = \begin{pmatrix} 1 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 1 & 0 & \cdots & 0 \\ 0 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & 0 \end{pmatrix} \chi$$

w/ vacuum energy $V \sim -O(\frac{m}{16\pi^2})^4$.

Another minimum on another direction $M^{ij} = \phi \delta^{ij}$. All dual quarks are massive and can be integrated out, yielding a pure $SU(N_f - N_c)$ supersymmetric Yang–Mills theory that leads to a gaugino condensate. Effective superpotential:

$$W = (N_f - N_c) \left(\frac{\kappa^{N_f} \det M}{\Lambda^3 N_c - 2N_f} \right)^{1/(N_f - N_c)}.$$

The potential in ASQCD:

$$V = N_f \Lambda^4 \left| \frac{\kappa \phi}{\Lambda} \right|^{2N_c/(N_f - N_c)} - (2N_f - 3N_c) m \Lambda^3 \left(\frac{\kappa \phi}{\Lambda} \right)^{N_f/(N_f - N_c)} + c.c.$$

Minimum at

$$\phi = \kappa^{-1} \Lambda \left(\frac{2N_f - 3N_c}{N_c} \frac{m}{\Lambda} \right)^{(N_f - N_c)/(2N_c - N_f)} \ll \Lambda$$

The mass for mesons from AMSB is loop suppressed and hence can be ignored.
The massless spectrum is once again the NGBs of $SU(N_f)_Q \times SU(N_f)_{\tilde{Q}} / SU(N_f)_V$.
There is no sign of phase transition as m is increased beyond Λ. [2104.01179]

5. **Apply this stabilization to electromagentic correction to pion mass (research in progress Y.Lin, H.Murayama)**

Toy model:

I first consider a toy model that has $SO(3)$ symmetry where $SO(2) = U(1)$ subgroup is gauged. We have three chiral superfields $\phi_i (i = 1,2,3)$ and X with the superpotential

$$W = \lambda X (\phi_i \phi_i - v^2).$$

Without supersymmetry breaking, we find the minimum of the potential

$$V = \left| \frac{\partial W}{\partial X} \right|^2 + \sum_i \left| \frac{\partial W}{\partial \phi_i} \right|^2 = |\lambda(\phi_i \phi_i - v^2)|^2 + \sum_i |2\lambda X \phi_i|^2 + \frac{e^2}{2} (-i\phi_2^* \phi_3 + i\phi_3^* \phi_2)^2$$

to be

$$\phi_1 = v, \phi_2 = \phi_3 = 0, X = 0$$

X and ϕ_1 together acquire mass $2\lambda v$, while $\phi_{2,3}$ are massless.

Tree-level AMSB effects are given by $-m\left(\varphi_i \frac{\partial W}{\partial \varphi_i} - 3W\right) + c.c.$, in our case

$$V_{\text{AMSB}} = -2m\lambda X v^2 + c.c.$$

To the leading order in m, the minimum shifts and X acquires an expectation value

$$V(X) = |2\lambda X v|^2 - 2m\lambda X v^2 + c.c.$$

$$\frac{\partial V}{\partial X^*} = 4|\phi_i|^2 v^2 X - 2m\lambda v^2 = 0 \Rightarrow X = \frac{m}{2\lambda}$$

The whole potential is

$$V = |\lambda(\varphi_i\varphi_i - v^2)|^2 + \sum_i |2\lambda X \varphi_i|^2 + \frac{e^2}{2}(-i\varphi_2^*\varphi_3 + i\varphi_3^*\varphi_2)^2 - 2m\lambda X v^2 + c.c.$$

pseudoscalars in $\phi_{2,3}$ remain massless, while scalars acquire mass. The D-term potential $\frac{e^2}{2}(-i\phi_2^*\phi_3 + i\phi_3^*\phi_2)^2$ does not give a mass unless $\phi_{2,3}$ acquire VEVs breaking $U(1)$. We ignore it for the discussion below.

$$\frac{\partial V}{\partial \phi_1^*} = 2\lambda\phi_1^*\lambda(\phi_i\phi_i - v^2) + 4\lambda^2 \mid X \mid^2 \phi_i = 0 \Rightarrow \lambda^2(\phi_i\phi_i - v^2) = -2\lambda^2 \mid X \mid^2 = \frac{-1}{2}m^2$$

$$(\phi_2^* \ \phi_2)\begin{pmatrix} 4\lambda^2|X|^2 & 2\lambda^2(\phi_1^2 - v^2) \\ 2\lambda^2(\phi_1^2 - v^2) & 4\lambda^2|X|^2 \end{pmatrix}\begin{pmatrix}\phi_2 \\ \phi_2^*\end{pmatrix} = (\phi_2^* \ \phi_2)\begin{pmatrix} m^2 & -m^2 \\ -m^2 & m^2 \end{pmatrix}\begin{pmatrix}\phi_2 \\ \phi_2^*\end{pmatrix}$$

Eigenvalues are $2m^2$ and 0. As expected.

Now we include one-loop γ and two-loop m^2. Naively $m_{2,3}^2$ for $\phi_{2,3}$ are negative (tachyonic). It would be interesting to see how the minimization fixes this problem.

$$\mathcal{L} = \int d^4\theta \mathcal{E}^*\mathcal{E}\mathcal{Z}\left(\frac{\mu}{M\mathcal{E}}\right)\phi_i^*\phi_i + \int d^2\theta \mathcal{E}^3 W(\phi) + c.c.$$

w/ $\mathcal{E} = 1 + m\theta^2$.

Rescale $\phi \to \phi/\mathcal{E}$

$$\mathcal{L} = \int d^4\theta Z_i(\left(1 - \frac{1}{2}\theta^2 m\gamma_i\right)\left(1 - \frac{1}{2}\bar{\theta}^2 m^*\gamma_i\right) + \frac{1}{4}\theta^2 m\bar{\theta}^2 m^*\dot{\gamma}_i)\phi_i^*\phi_i + \int d^2\theta \mathcal{E}^3 W(\phi_i/\mathcal{E}) + c.c.$$

Further rescale $\phi_i \to Z_i^{-1/2}\phi_i(1 + \frac{1}{2}\theta^2 m\gamma_i)$, gives the potential

$$V = Z_i^{-1}W_i^*W_i - \frac{1}{4}mm^*\dot{\gamma}\phi_i^*\phi_i - (3mW - m\left(1 - \frac{1}{2}\gamma_i\right)\phi_i W_i + c.c.)$$

where $W = \lambda X(\phi_i\phi_i - v^2)$.

Consider only the one-loop level AMSB and keep only X and ϕ_1

$$V = |\lambda(\phi_1\phi_1 - v^2)|^2 + \mid 2\lambda X\phi_1 \mid^2 + m\lambda X\big((\gamma_X + 2\gamma_1)\phi_1\phi_1 - (2 + \gamma_X)v^2 + c.c.\big).$$

One gives the mass correction up to one-loop by giving the eigenvalues

$$-2\mathrm{m}^2(\mathrm{t}\gamma\phi_1 - \mathrm{t}\gamma\phi_2), -2\mathrm{m}^2(-1 + \mathrm{t}\gamma\phi_1 + \mathrm{t}\gamma\phi_2)$$

where AMSB loop-level factors

$$\gamma_X = \frac{-1}{8\pi^2} 3\lambda^2$$
$$\gamma_1 = \frac{-1}{8\pi^2} \lambda^2$$
$$\gamma_2 = \frac{1}{8\pi^2} (2e^2 - \lambda^2)$$

thus eigenvalues (mass)

$$\frac{\mathrm{e}^2\mathrm{m}^2\mathrm{t}}{2\pi^2}, \frac{1}{2}\mathrm{m}^2\left(4 + \frac{\mathrm{t}(-\mathrm{e}^2 + \lambda^2)}{\pi^2}\right).$$

It works.

[I think I shouldn't include more of this chapter without the agreement from my collaborator since it has not published yet.]

ADS:

...

Free magnetic:

...